A HAND-CHECKABLE PROOF THAT $R(B_{66}, B_{67}) = 267$

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Abstract. Chung, Du and Wesley showed that $R(B_{66}, B_{67}) = 267$, the case $n = 67$ of Wesley's conjecture $R(B_{n-1}, B_n) = 4n - 1$, by a 266-vertex certificate. We describe their graph as a two-block circulant graph over $\mathbb{Z}_{133}$ invariant under the cube roots of unity, which reduces the proof to ten cases checkable by hand, and verify it in Lean 4.

1. Introduction

The book B_k is the graph $K_2 + \overline{K_k}$: k triangles sharing one edge, its spine. For graphs G and H, the Ramsey number $R(G, H)$ is the least N such that every graph on N vertices contains G or has a complement containing H. Rousseau and Sheehan [9] proved, from Goodman's count of monochromatic triangles [6], that

$$R(B_{n-1}, B_n) \le 4n - 1 \qquad (n \ge 2). \tag{1.1}$$

Wesley [12] proved equality for $n \le 20$ and whenever $2n - 1$ is a prime power congruent to 1 modulo 4, and conjectured equality for every n. The conjecture is one of the open problems of Epoch AI's FrontierMath collection [5], which records further progress. Turturean [11] constructed extremal graphs for all $n \le 56$ and for fifteen further values below 100, partly through a cyclotomic construction of Szekeres [10] that applies when $4n-1$ is a prime congruent to 3 modulo 8; Chung, Du and Wesley [3] found three further infinite families, from conference graphs, from Legendre pairs and from Yamada–Pott designs; and Price obtained $n = 70$ [5].[1] The value $n = 67$ is not covered by these families, but the public data accompanying [3] settle it by a finite certificate [4, Section 5.1]: a 266-vertex graph, described there as a regular two-fibre colouring built from multiplier orbits over $\mathbb{Z}_{133}$, in which every red edge has at most 65 common red neighbours and every blue edge at most 66 common blue neighbours.

Theorem (Chung, Du and Wesley [4])**.** *$R(B_{66}, B_{67}) = 267$.*

We found the same graph independently, by the computer search described in Section 4, before we knew of [4]; the two graphs are isomorphic (Remark 4.2). This note gives an explicit description of it and a proof that does not rest on a computer. The upper bound is (1.1). For the lower bound the graph is a two-block circulant graph over $\mathbb{Z}_{133}$, the shape used by Wesley [12] and Turturean [11], and its connection sets are invariant under multiplication by the nine cube roots of unity modulo 133. Section 2 reduces the two book conditions to one inequality (2.1) on the connection sets; the symmetry then reduces (2.1) to ten cases, which Table 1 records. Every step can be checked by hand, and the graph has also been checked by

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[1] The values $n \le 100$ that none of the constructions cited here covers are 59, 60, 64, 65, 68, 72, 73, 76, 78, 80, 81, 84, 88, 89, 92, 93, 94, 96 and 100; the value $n = 58$ is settled in [2]. This list is our reading of [3, 4, 5, 11, 12] as of 15 October 2026; apart from $n = 58$ it agrees with the ledger in [4, Section 6.1].

two independent programs and in Lean 4 (see the end of the paper); [4] checks its certificate by a Python program.

2. Two-block circulant graphs

Throughout, $n \geq 2$, $v = 2n-1$ and $G = \mathbb{Z}_v$. For $X, Y \subseteq G$ and $d \in G$ put

$$\mathrm{N}_{X,Y}(d) = |X \cap (Y+d)| = \#\{y \in Y : \ y+d \in X\}.$$

Translating by $-d$ shows $\mathrm{N}_{X,X}(-d) = \mathrm{N}_{X,X}(d)$, and $\sum_{d \in G} \mathrm{N}_{X,Y}(d) = |X|\,|Y|$.

Let $A \subseteq G \setminus \{0\}$ with $A = -A$ and $|A| = n-1$, let $C = G \setminus (A \cup \{0\})$, so that $C = -C$ and $|C| = n-1$, and let $B \subseteq G$ with $|B| = n-1$. The graph $\Gamma = \Gamma(A,B)$ has vertex set $G \times \{0,1\}$, and

$$(x,0) \sim (y,0) \iff y-x \in A, \qquad (x,1) \sim (y,1) \iff y-x \in C,$$

and $(x,0) \sim (y,1)$ if and only if $y - x \in B$. Since A and C are symmetric and avoid 0, Γ is a simple graph; in the notation of [12] it is $\Gamma_G(D_{11}, D_{12}, D_{22})$ with $D_{11} = A$, $D_{12} = B$, $D_{22} = C$. A vertex $(x,0)$ has $|A| + |B| = 2n-2$ neighbours and a vertex $(x,1)$ has $|C| + |B| = 2n-2$, so Γ is $(2n-2)$-regular on $4n-2$ vertices. The translations $(x,i) \mapsto (x+t, i)$ are automorphisms. Write $\lambda(p,q)$ for the number of common neighbours of two vertices, $[\,P\,]$ for the indicator of a statement P, and

$$f(d) = \mathrm{N}_{A,A}(d) + \mathrm{N}_{B,B}(d) \qquad (d \in G).$$

Lemma 2.1. *For $d \neq 0$,*

$$\lambda\big((0,0),(d,0)\big) = f(d), \qquad \lambda\big((0,1),(d,1)\big) = f(d) - 1 + 2[\,d \in A\,],$$

and for every $d \in G$, $\lambda\big((0,0),(d,1)\big) = n - 1 - [\,d \in B\,]$.

Proof. A vertex $(z,0)$ is a common neighbour of $(0,0)$ and $(d,0)$ when $z \in A$ and $z - d \in A$, and $(z,1)$ is one when $z \in B$ and $z - d \in B$; this gives $\mathrm{N}_{A,A}(d) + \mathrm{N}_{B,B}(d)$.

For $(0,1)$ and $(d,1)$, the common neighbours $(z,1)$ are counted by $\mathrm{N}_{C,C}(d)$. A vertex $(z,0)$ is adjacent to $(0,1)$ when $-z \in B$ and to $(d,1)$ when $d - z \in B$; with $w = -z$ these say $w \in B$ and $w + d \in B$, so there are $\mathrm{N}_{B,B}(-d) = \mathrm{N}_{B,B}(d)$ of them. To compute $\mathrm{N}_{C,C}(d)$, let $S = A \cup \{0\}$, so $|S| = n$ and $C = G \setminus S$. Then $|C \cap (C+d)| = v - |S \cup (S+d)| = v - 2n + |S \cap (S+d)|$. The set $S \cap (S+d)$ is the disjoint union of $A \cap (A+d)$, of $\{0\}$ when $-d \in A$, and of $\{d\}$ when $d \in A$ (neither 0 nor d lies in $A \cap (A+d)$, because $0 \notin A$). As $A = -A$ and $v - 2n = -1$, we get $\mathrm{N}_{C,C}(d) = \mathrm{N}_{A,A}(d) - 1 + 2[\,d \in A\,]$.

For $(0,0)$ and $(d,1)$, a vertex $(z,0)$ is a common neighbour when $z \in A$ and $d - z \in B$, that is, $z = d - b$ with $b \in B$ and $d - b \in A$; since $A = -A$ the last condition reads $b - d \in A$. A vertex $(z,1)$ is a common neighbour when $z \in B$ and $z - d \in C$. Together, $\lambda = \#\{b \in B : \ b - d \in A \cup C\} = \#\{b \in B : \ b \neq d\} = n - 1 - [\,d \in B\,]$. □

Lemma 2.2. *In a $(2n-2)$-regular graph on $4n-2$ vertices, two distinct non-adjacent vertices p, q have exactly $\lambda(p,q)$ common non-neighbours other than p and q.*

Proof. The neighbourhoods of p and q lie in the other $4n-4$ vertices and their union has $2(2n-2) - \lambda(p,q)$ elements; the remaining $\lambda(p,q)$ vertices are adjacent to neither. □

Proposition 2.3. *$\Gamma(A,B)$ contains no B_{n-1} and its complement contains no B_n if and only if*

$$f(d) \leq n - 1 - [\,d \in A\,] \qquad \text{for every } d \in G \setminus \{0\}. \tag{2.1}$$

In that case $R(B_{n-1}, B_n) = 4n-1$.

Proof. An edge in k triangles is the spine of a B_k. So Γ contains no B_{n-1} if and only if every edge has at most $n-2$ common neighbours, and by Lemma 2.2 the complement contains no B_n if and only if every pair of distinct non-adjacent vertices has at most $n-1$ common neighbours. Both conditions together say $\lambda(p,q) \le n-1-[\,p \sim q\,]$ for all $p \ne q$, and by translation it suffices to take $p = (0,i)$. We use Lemma 2.1. For $p = (0,0)$, $q = (d,1)$ we have $p \sim q$ if and only if $d \in B$, so the condition holds with equality; the pairs $(0,1), (d,0)$ are translates of these. For $p = (0,0)$, $q = (d,0)$ with $d \ne 0$ we have $p \sim q$ if and only if $d \in A$, and the condition is (2.1). For $p = (0,1)$, $q = (d,1)$ we have $p \sim q$ if and only if $d \notin A$, and the condition $f(d) - 1 + 2[\,d \in A\,] \le n - 2 + [\,d \in A\,]$ is again (2.1). The last sentence follows from (1.1), since Γ has $4n-2$ vertices. □

Remark 2.4. Summing over $d \ne 0$ gives $\sum f(d) = 2(n-1)(n-2)$ and $\sum (n-1-[\,d \in A\,]) = (n-1)(2n-3)$. So when (2.1) holds, the slacks $s(d) = n-1-[\,d \in A\,] - f(d)$ are non-negative integers with sum $n-1$, and (2.1) is an equality for at least $n-1$ of the $2n-2$ values of d. The cross edges never matter: B enters only through its autocorrelation $\mathrm{N}_{B,B}$. For the Paley graphs of [12] (v a prime power, $A = B$ the nonzero squares), $s = 1$ on A and $s = 0$ on C.

Lemma 2.5. *If $u \in \mathbb{Z}_v^\times$ satisfies $uA = A$ and $uB = B$, then $f(ud) = f(d)$ and $[\,ud \in A\,] = [\,d \in A\,]$; moreover $f(-d) = f(d)$ and $[\,-d \in A\,] = [\,d \in A\,]$. Hence, if $H \le \mathbb{Z}_v^\times$ fixes A and B, it suffices to check (2.1) for one d in each orbit of the group $\pm H$ on $G \setminus \{0\}$.*

Proof. The map $y \mapsto uy$ is a bijection from $X \cap (Y+d)$ onto $uX \cap (uY + ud)$, so $\mathrm{N}_{uX,uY}(ud) = \mathrm{N}_{X,Y}(d)$. The statements about $-d$ follow from $\mathrm{N}_{X,X}(-d) = \mathrm{N}_{X,X}(d)$ and $A = -A$. □

3. The case $n = 67$

Let $n = 67$ and $v = 133 = 7 \cdot 19$. Let

$$H = \{u \in \mathbb{Z}_{133} : u^3 = 1\} = \{1, 11, 30, 39, 58, 64, 102, 106, 121\},$$

the group of cube roots of unity; under $\mathbb{Z}_{133} \cong \mathbb{Z}_7 \times \mathbb{Z}_{19}$ it is $\{1,2,4\} \times \{1,7,11\}$. Since $|H|$ is odd, $-1 \notin H$ and $\pm H$ has order 18. Its orbits on $\mathbb{Z}_{133} \setminus \{0\}$ are the following ten: the 108 units form six orbits of size 18, with representatives $1, 2, 3, 4, 6, 8$; on the 18 nonzero multiples of 7 the group acts through its image $\pm\{1,7,11\}$ modulo 19, which has order 6, giving three orbits, with representatives $7, 14, 28$; and on the six nonzero multiples of 19 it acts through its image $\pm\{1,2,4\} = \mathbb{Z}_7^\times$ modulo 7, giving one orbit, with representative 19. Put

$$A = \pm H \cdot \{1, 3, 4, 14, 28\},$$
$$B = H \cdot \{2, 5, 8, 10, 12, 14, 20, 28, 57, 70\}.$$

The first is a union of three orbits of size 18 and two of size 6, so $|A| = 66$ and $A = -A$; the second is a union of six H-orbits of size 9 and four of size 3, so $|B| = 66$. Explicitly,

$$\begin{aligned} A = \{&1, 3, 4, 9, 10, 11, 12, 13, 14, 16, 21, 23, 25, 27, 28, 30, 31, 33, 34, 35, 36, 39, 40,\\ &41, 42, 43, 44, 48, 52, 58, 59, 63, 64, 69, 70, 74, 75, 81, 85, 89, 90, 91, 92, 93, 94,\\ &97, 98, 99, 100, 102, 103, 105, 106, 108, 110, 112, 117, 119, 120, 121, 122, 123,\\ &124, 129, 130, 132\},\\ B = \{&2, 5, 8, 10, 12, 13, 14, 17, 18, 20, 21, 22, 24, 26, 27, 28, 31, 34, 37, 42, 45, 46, 48,\\ &50, 54, 55, 57, 60, 62, 63, 65, 68, 69, 70, 71, 73, 75, 78, 79, 83, 87, 88, 89, 91, 94,\\ &95, 96, 98, 103, 105, 107, 108, 109, 110, 111, 113, 114, 115, 116, 122, 124, 125,\\ &128, 129, 131, 132\}. \end{aligned}$$

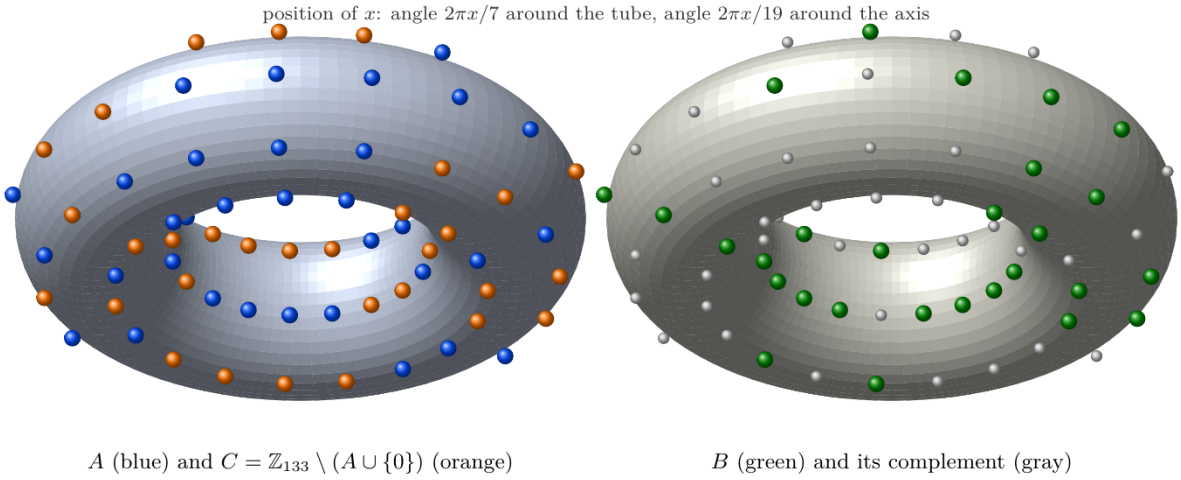


FIGURE 1. The sets A and B for $n = 67$. The residue $x \in \mathbb{Z}_{133}$ sits at angle $2\pi x/7$ around the tube and $2\pi x/19$ around the axis, so multiplication by $H = \{1, 2, 4\} \times \{1, 7, 11\}$ permutes the points of each picture within its colours.

TABLE 1. The inequality (2.1) for $n = 67$ on the ten orbits of $\pm H$. The last column is the slack $s(d)$ of Remark 2.4; the sizes times the slacks add up to 66.

d	$\lvert \pm Hd \rvert$	$d \in A$	$\mathrm{N}_{A,A}(d)$	$\mathrm{N}_{B,B}(d)$	$f(d)$	$66 - [\,d \in A\,]$	$s(d)$
1	18	yes	36	28	64	65	1
2	18	no	35	31	66	66	0
3	18	yes	30	35	65	65	0
4	18	yes	32	32	64	65	1
6	18	no	33	33	66	66	0
8	18	no	31	35	66	66	0
7	6	no	33	32	65	66	1
14	6	yes	30	33	63	65	2
28	6	yes	31	34	65	65	0
19	6	no	30	34	64	66	2

The set B is fixed by H but not by -1: it meets $-B$ in 42 elements. Figure 1 draws both sets on the torus $\mathbb{Z}_7 \times \mathbb{Z}_{19}$.

Theorem 3.1. *The sets A and B above satisfy (2.1) for $n = 67$. Consequently $\Gamma(A, B)$ is a 132-regular graph on 266 vertices with no B_{66} whose complement has no B_{67}, and $R(B_{66}, B_{67}) = 267$.*

Proof. An element u of H fixes B, because B is a union of H-orbits, and fixes A, because A is a union of $\pm H$-orbits. By Lemma 2.5 it suffices to check (2.1) at the ten representatives, and Table 1 lists $\mathrm{N}_{A,A}(d)$ and $\mathrm{N}_{B,B}(d)$ there; each entry is the number of elements of a 66-element

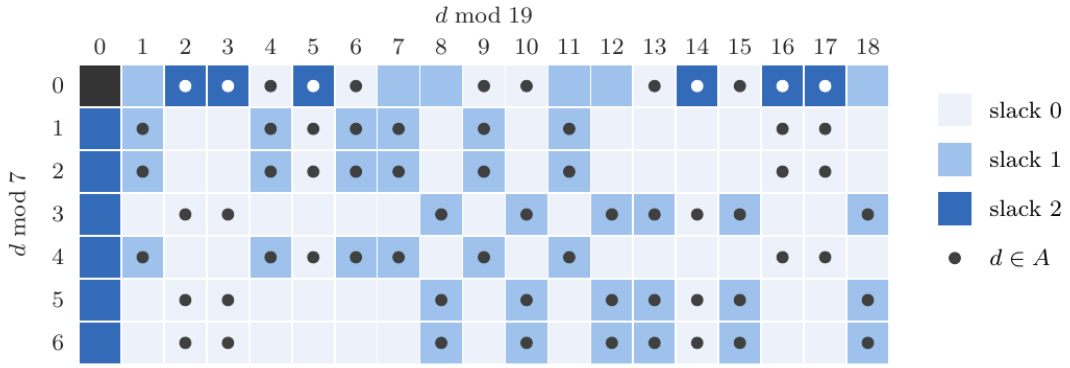


FIGURE 2. The slack $s(d) = 66 - [\,d \in A\,] - f(d)$ for $n = 67$ at every $d \neq 0$, with d placed by its residues modulo 7 and 19; the black cell is $d = 0$, and a dot marks $d \in A$. Equality holds for 78 values of d, the slack is 1 for 42 and 2 for 12.

set whose translate by $-d$ stays in the set.[2] In every row $f(d) \leq 66 - [\,d \in A\,]$. Proposition 2.3 gives the rest. □

4. REMARKS

Remark 4.1. The function f is not constant on A or on C (Figure 2): it takes the values 63, 64, 65 on A and 64, 65, 66 on C. So the example is not a difference-set construction of the kind behind the infinite families in [10, 11, 12], where f is constant on A and on C.

Remark 4.2. The graph $\Gamma(A, B)$ is isomorphic to the certificate `finite_n67` of [4], the 266×266 matrix with SHA-256 digest `819dbb31...a1cf1` stated there. Both graphs are 132-regular, have the same numbers of edges in 63, 64 and 65 triangles (798, 2793 and 13965) and the same spectrum, and an explicit isomorphism, found by the VF2++ algorithm, is archived in [1]: relabelling $\Gamma(A, B)$ by it gives a matrix with the published digest, and it agrees entry by entry with the matrix of [4].

Remark 4.3. We found the example by a computer search. For fixed $v = 2n-1$ and a subgroup $H \leq \mathbb{Z}_v^\times$, the sets A and B are taken to be unions of H-orbits, with $A = -A$; one 0/1 variable for each orbit in A and one for each orbit in B turn (2.1) into a system of quadratic inequalities, one for each H-orbit of d, together with $|A| = |B| = n - 1$. For $v = 133$ the cube roots of unity have 20 orbits on the nonzero residues, and the CP-SAT solver of OR-Tools [8] finds a solution in under a second. The same search did not produce examples for the other values in the footnote to the introduction, for the subgroups and the time limits we tried; some of these systems are infeasible. No step of the proof above depends on the search.

Question 4.4. Is there an infinite family of two-block circulant graphs, with connection sets invariant under a large multiplier group but f not constant on A, that settles further cases of $R(B_{n-1}, B_n) = 4n - 1$? The smallest open case is now $n = 59$, with $v = 117$. For $n = 58$ the graph $\Gamma(A, B)$ itself does not exist, since a symmetric set A avoiding 0 has even size while $|A| = n - 1 = 57$; that case is settled in [2] by a graph of the shape Turturean uses for even

[2]For instance $\mathrm{N}_{A,A}(1) = 36$ is the number of pairs $a - 1, a$ of consecutive residues in A: the runs $3, 4$; $9, \ldots, 14$; $27, 28$; $30, 31$; $33, \ldots, 36$; $39, \ldots, 44$; $58, 59$; $63, 64$; $69, 70$; $74, 75$; $89, \ldots, 94$; $97, \ldots, 100$; $102, 103$; $105, 106$; $119, \ldots, 124$; $129, 130$ contribute $1 + 5 + 1 + 1 + 3 + 5 + 1 + 1 + 1 + 1 + 5 + 3 + 1 + 1 + 5 + 1 = 36$.

n [11], with $|A| = n$, $|B| = n-1$ and the third block the complement of $A \cup \{0\}$, and that graph has no multiplier symmetry.

Formal verification. Theorem 3.1 and the equality $R(B_{66}, B_{67}) = 267$ are also proved in Lean 4 [7], without any library beyond the Lean core. For the lower bound the development defines the graph $\Gamma(A, B)$ on the vertices $0, \dots, 265$ by the two bit masks of A and B, and the kernel evaluates the common neighbours and common non-neighbours of all 35,245 pairs of vertices and checks that the adjacency is symmetric: every edge has at most 65 common neighbours and every pair of distinct non-adjacent vertices has at most 66 common non-neighbours. For the upper bound (1.1) the development proves Goodman's count in the form used by Rousseau and Sheehan: if every edge of a graph on N vertices lies in at most a triangles and every non-edge has at most b common non-neighbours, then counting the ordered triples of distinct vertices by the colours of their three pairs gives

$$N(N-1)(N-2) \le \sum_{w} \bigl(3\, d(w)\, \bar{d}(w) + a\, d(w) + b\, \bar{d}(w)\bigr),$$

where $d(w)$ is the degree of w and $\bar{d}(w) = N - 1 - d(w)$. For $N = 267$, $a = 65$ and $b = 66$ each term on the right is at most $266 \cdot 265$, with equality only when $d(w) = 133$, and then the degrees would add up to the odd number $267 \cdot 133$. The final theorem states that 267 is the least N such that every graph on N vertices contains B_{66} or has a complement containing B_{67}; it depends only on the standard axioms `propext`, `Classical.choice` and `Quot.sound`. The graph is also checked by two independent programs, in Python and in C, which read the sets A and B and test all pairs directly. The development, the programs and the search are archived in [1].

Use of generative AI. The author used Claude (Anthropic) to search for the example, to check it by computer and in Lean, and to draft the text; the author verified every step and is responsible for the content.

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